Università degli Studi di Napoli Federico II

Corso di Laurea Magistrale in Fisica

# Canonical BRST Quantization of the Gravitational Field

Candidato:
Luigi Petillo,
lu.petillo@studenti.unina.it
Matricola N94000914

Supervisor:
Prof. Wolfgang Mück,
wolfgang.mueck@unina.it

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**Summary**

Fundamental physical theories are generally formulated as gauge theories: their mathematical description employs more variables than the actual physical degrees of freedom. This redundancy is managed through gauge transformations: the observable physical content is defined strictly by the quantities that remain invariant under these transformations.
This work begins with a preliminary study of gauge theories, as widely presented in [1], within the Lagrangian framework, where the vanishing of the Hessian determinant prescribes the system's redundancy. Moving to the Hamiltonian formalism, some defining equations for the conjugate momenta emerge as constraints, classifying gauge systems as constrained Hamiltonian systems. Following the extension of the Hamiltonian and Dirac's classification of constraints—primary and secondary, first and second class— Dirac's conjecture posits that gauge transformations are generated solely by first-class constraints, a statement that holds true for virtually all systems of physical relevance. Furthermore, it is demonstrated that in a general covariant kinematic description, introducing a non-physical evolution parameter leads to a Zero Hamiltonian condition, as occurs in General Relativity (GR), rendering observables invariant under time reparameterization. A quantum analogue of these statements will subsequently be recovered.
The study proceeds with the general Becchi-Rouet-Stora-Tyutin (BRST) formalism as presented in [2] and [3]. After analyzing standard examples, such as Maxwell's theory—where the kinetic operator is non-invertible due to gauge redundancy—an outline of the Faddeev-Popov procedure is given to derive the extended BRST-invariant action. BRST is a general method for quantizing theories with first-class constraints, i.e., with gauge symmetries. Although there exists an independent Hamiltonian approach, pedagogically, BRST is easier introduced starting from the path integral. Indeed, the ghost fields arise from the formal expression of the Faddeev-Popov determinant as a path integral over fermionic fields. By promoting local gauge symmetry to a global BRST symmetry, Noether's theorem is applied to compute the nilpotent BRST charge, which acts as the generator for the BRST transformations of the fields, including ghosts. Within this framework, the treatment of constraints is notably more elegant: physical states are proven to be annihilated by both the BRST charge and the ghost charge. Consequently, the physical Hilbert space is rigorously defined as the cohomology of the BRST charge.
Once the formalism is established, the BRST treatment is explicitly applied to scalar QED. This instructive example clearly exhibits two primary constraints, which are utilized to eliminate two (Hamiltonian) dof. Following Canonical Quantization, the Hamiltonian is computed, and the Heisenberg relations are verified. The BRST and ghost charges are derived, confirming that they correctly generate their respective transformations for all fields.
Finally, this comprehensive methodology is applied to Gravity. The BRST action for gravity is constructed based on the work of Kugo and Ojima [4]. The Einstein-Hilbert action is split into two components: a bulk term, denoted as $S_{\text{grav}}$, containing only first derivatives of the metric, and a remaining boundary term. Because the variation of $S_{\text{grav}}$ contains no terms proportional to the variation of metric derivatives and correctly yields the vacuum Einstein equations under Hamilton's principle, it is adopted as the starting action, expressed in terms of the inverse metric density $\tilde{g}^{\mu\nu} \equiv \sqrt{-g}g^{\mu\nu}$, chosen as Lagrangian variable. The gauge-fixing condition $\partial_\mu \tilde{g}^{\mu\nu} = \mathcal{F}^\nu$ is implemented, with $\mathcal{F}^\nu$ acting as an external field. The BRST varia-

tion of this newly constructed action results in a boundary term, which is crucial for evaluating the Noether current. After computing the current, an improvement is selected to eliminate second derivatives of the fields, making the subsequent Hamiltonian treatment straightforward. All computations are performed explicitly, both analytically and, where necessary, using Cadabra2 integrated with Python. The transition to the Hamiltonian formalism is performed by employing time slicing and the ADM formalism of Canonical Gravity. However, inspired by [5], an original choice of canonical variables is adopted in place of the standard Lapse and Shift fields. The computation of conjugate momenta reveals eight primary constraints, utilized to eliminate eight (Hamiltonian) dof. The Hamiltonian is thereby constructed, and canonical quantization is performed. The same verifications applied to scalar QED are conducted: the validity of the Heisenberg relations (which correctly yield the gauge-fixing equation for auxiliary fields) and the proper generation of BRST transformations by the charge.
From the expression of the BRST charge density, it emerges that the bulk Hamiltonian can be elegantly written as BRST-exact, both for $\mathcal{F}^\nu = 0$ and $\mathcal{F}^\nu \neq 0$, through an additive modification to the gauge-fixing fermion (basically given by the momentum conjugate to the ghost field $c^0$). This property has profound consequences, as hinted in [6]. The bulk Hamiltonian exhibits vanishing matrix elements between physical states, meaning the Heisenberg evolution of physical amplitudes is entirely driven by the boundary Hamiltonian. However, the bulk Hamiltonian significantly contributes to the evolution of gauge-variant operators: the bulk Hamiltonian flow acts as a time reparameterization, consistently mirroring the classical behavior of all general covariant systems. Similarly, the BRST cohomology shows many interesting features. For example, starting from a fixed would-be-classical background as vacuum, there exists a class of equivalent physical states, built as coherent states: the metric expectation values evaluated on them match exactly those obtained classically by performing a gauge transformation on the starting metric.

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[3] M. D. Schwartz, "Quantum Field Theory and the Standard Model", Cambridge University Press (2013).

[4] T. Kugo and I. Ojima, "Subsidiary Conditions and Physical S-matrix Unitarity in indefinite-metric Quantum Gravitation theory", Nuclear Physics B144, pp. 234-252 (1978).

[5] L. Berezhiani, G. Dvali and O. Sakhelashvili, "Consistent Canonical Quantization of Gravity: Recovery of Classical GR from BRST-invariant Coherent States", arXiv:2409.18777 [hep-th] (2024).

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